GENESIS: Survival-Only Learning Agents in a Seasonal Foraging Ecology

A testbed, baselines, and a behavioural anatomy of the first learners

Heet Ranpura* **Advik Divekar**

October 2026

Abstract

We ask what individual competence and social organisation emerge when minimally specified learning agents are rewarded *only* for keeping their own bodies alive. We introduce GENESIS, a fast, deterministic multi-agent foraging ecology: a procedurally generated 256×256 island with rivers, seasons, weather, wildfire, floods, famine, and food-limited prey and predator populations, in which every agent's reward is the reduction of its own homeostatic drive (energy, hydration, health, warmth) and a death penalty. Social behaviour is never rewarded; it is possible (food can be dropped and picked up; large prey need two hunters) but must emerge. We train a single parameter-shared recurrent PPO policy for 256 bodies (19.7M agent-steps) and evaluate it, with hypotheses registered in advance, on five held-out worlds against random, greedy and hand-written heuristic foragers. Learners cut mortality by more than half relative to random and greedy agents (memoryless 1.50 [1.30, 1.75], GRU 1.67 [1.45, 1.92] deaths per agent-year vs. random 3.83 [3.53, 4.19]) and match, but do not beat, the heuristic (1.50 [1.04, 2.11]); the result replicates across two independent implementations. A behavioural analysis of every agent at every tick shows that learners discover *graded homeostatic control* (the probability of drinking rises from 0.09 when hydrated to 0.52 when parched), *riverside sedentism* (median 2.0 cells from fresh water and 14-cell home ranges, vs. 3.6 and 38 for the heuristic), and a critic whose value estimate tracks the body's distance from its set-point ($r = -0.65$ to -0.75). They never learn to hunt and die of starvation in autumn and winter (76–77% of deaths). Diet probes show this is a local optimum, not a property of the world: plants alone cannot cover basal metabolism, while a hunting forager breaks even. Learners appear clustered, but the clustering is fully explained by shared place preferences (nearest-neighbour ratio 0.84 against a habitat null, 0.98–1.01 against a place-preference null): as preregistered, no social behaviour has emerged yet.

1 Introduction

Human sociality arose under survival pressure: finding water, surviving winter, hunting animals too large to take alone. Whether comparable organisation arises in artificial agents that face similar pressures, *without being told to cooperate*, is a question about which parts of human social life are convergent outcomes of ecology and which are contingent (Duéñez-Guzmán et al., 2023). Agent-based models showed long ago that simple rules can generate trade, migration and inequality (Epstein and Axtell, 1996), and deep multi-agent reinforcement learning has produced tool use (Baker et al., 2020), bartering (Johanson et al., 2022), cooperation in social dilemmas (Leibo et al., 2017, 2021; Perolat et al., 2017) and emergent signalling (Foerster et al., 2016; Lazaridou et al., 2017). Most of these results, however, either hand-code behaviour or shape the reward (team rewards, explicit social or communication objectives), which makes it hard to say what the *ecology alone* produces.

GENESIS is built to answer that narrower question with unusually strict rules: (i) the reward is the agent's own homeostasis (Keramati and Gutkin, 2014) and nothing else; (ii) capabilities such as

*Correspondence: heetranpura1@gmail.com. Code: https://github.com/HeetRanpura/Genesis.

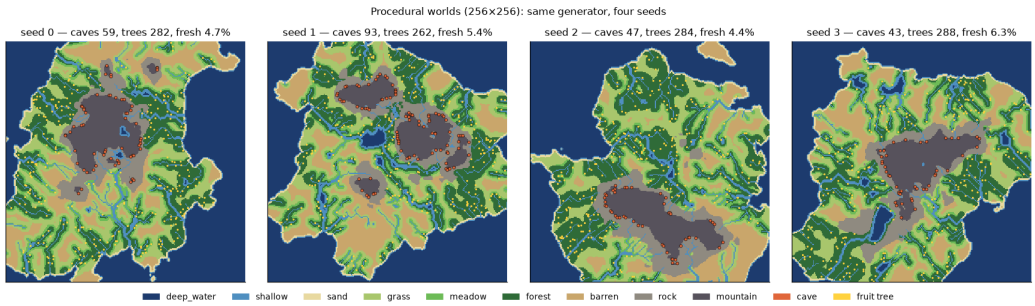


Figure 1: Four procedurally generated worlds (same generator, different seeds). Elevation from domain-warped fBm, lakes from priority-flood depression filling (Barnes et al., 2014), rivers from flow accumulation; the ocean is salt, so fresh water is scarce and spatially structured. Red: caves; yellow: fruit trees.

memory, communication, reproduction and building are switches, off by default; (iii) every run is seeded and bit-reproducible; (iv) non-learning baselines appear in every comparison; and (v) claims are made only on held-out worlds, with hypotheses written down before evaluation. This paper reports the first learning agents in that setting and, because a single mortality number hides *how* agents survive, a detailed behavioural anatomy of what they learned.

Contributions.

- **GENESIS**, an open, deterministic, Numba-accelerated ecology (≥100k environment steps per minute with full observations for 1,000 agents on a laptop) with food-limited fauna, a flight recorder that logs every agent at every tick, and exact replay (§3).
- **A preregistered evaluation** of parameter-shared recurrent PPO against three baselines on held-out worlds, replicated across two independent implementations (§5).
- **A behavioural anatomy** of the learners: time budgets, need-conditioned choice, space use, daily and seasonal rhythms, critic internals, individual differences, and an aggregation test with a place-preference null (§6).
- **An energy-economy probe** showing the learners' plant-only diet is a local optimum the world does not force (§7).

2 Related work

Artificial societies and artificial life. Sugarscape (Epstein and Axtell, 1996) grew social phenomena from hand-written rules; Polyworld (Yaeger, 1994) evolved neural foragers. Neural MMO (Suarez et al., 2019) provides a large survival world for many learning agents and is the closest modern platform; GENESIS differs in its strictly homeostatic reward, ecological dynamics (seasons, food-limited fauna, disturbances), held-out-world protocol and per-tick recording. Crafter (Hafner, 2022) studies single-agent survival skills. **Social learning in MARL.** Sequential social dilemmas (Leibo et al., 2017), common-pool resources (Perolat et al., 2017) and Melting Pot (Leibo et al., 2021) measure cooperation, usually with designed incentives; cultural transmission (Bhoopchand et al., 2023) and bartering (Johanson et al., 2022) emerge in purpose-built tasks. LLM-agent societies (Park et al., 2023) start from human cultural priors, which our agents lack by construction. **Homeostatic RL.** Defining reward as drive reduction toward physiological set-points (Keramati and Gutkin, 2014) grounds "survival" in a single, non-social objective.

3 The GENESIS environment

World. Each run generates a 256×256 island from its seed (Fig. 1) with ten terrain classes. A year has four eight-day seasons of 120 ticks per day (3,840 ticks per year). Weather follows a daily Markov

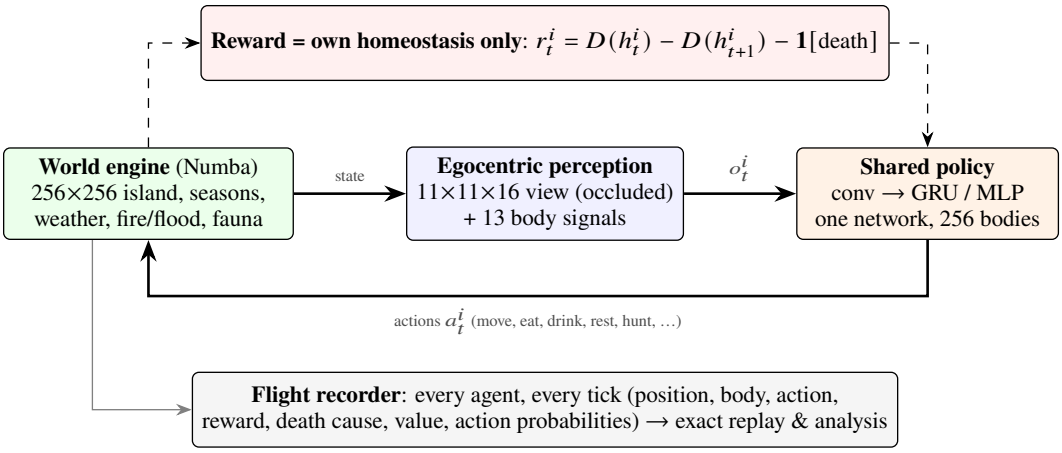


Figure 2: The GENESIS loop. One shared policy drives every body; each agent's reward depends only on its own homeostasis; every tick of every agent is recorded.

chain (rain, fog, drought, cold snaps, snow); rare disturbances (wildfire, spring floods, famine years) occur naturally and can be triggered. Plants regrow logistically toward a fertility map at seasonal rates; fruit trees bear in one season; small prey, large prey and predators are *food-limited*: they have energy stores, winter torpor and seasonal breeding, and graze, flee, herd and hunt. Population caps exist only as safety rails and never bind in our diagnostics. Each run starts after one year of ecological spin-up without agents.

Agents. Agents occupy one cell each and perceive an 11×11 egocentric patch with 16 channels (terrain, water, plants, fruit, prey, predators, other agents, fire, snow; line of sight blocked by mountains and dense forest; the radius shrinks at night and in fog), 13 interoceptive signals, the four nearest visible neighbours, and heard signals (unused here). Eleven actions: four moves, rest, eat, drink, hunt, pick up, drop, no-op. Bodies pay basal metabolism (5.0 energy units per year), movement by terrain, and dehydration (faster in heat); health falls when energy or water run out, when cold, in fire, or when attacked by predators. Large prey can be killed only by ≥2 attackers within six ticks, and carried food can be dropped where others can eat it. These features make cooperation *possible*; nothing rewards it.

Reward. For agent i with body $h^i \in [0,1]^4$ and set-point $h^* = (1, 1, 1, 0.8)$, the drive is $D(h) = \|\max(h^* - h, 0)\|_2$ and the reward is $r_t^i = D(h_t^i) - D(h_{t+1}^i) - \mathbf{1}[\text{death}]$ (Keramati and Gutkin, 2014). A unit test asserts that rewards depend only on the agent's own body.

Engineering. The simulator is written in Numba (Lam et al., 2015), is bit-reproducible across machines (all randomness is a counter-based hash of seed, tick, index and stream), exposes a PettingZoo interface (Terry et al., 2021), and runs one tick as a single fused kernel call. Observations are masked copies of an incrementally maintained observation image, which keeps 1,000 fully observing agents above 100k steps per minute on a fanless laptop. A flight recorder stores every agent's state, action, reward and death cause at every tick (and, for learners, the value estimate and action probabilities), and any run can be replayed bit-exactly.

4 Agents and protocol

Baselines. *Random* samples actions uniformly. *Greedy* attends to its lower of energy and hydration and walks to the nearest visible resource for it, else explores. *Heuristic* is a hand-written survivalist: it flees predators and fire, drinks and eats with hysteresis thresholds, hunts adjacent small prey, seeks shelter when cold, and rests at night. None has a social rule.

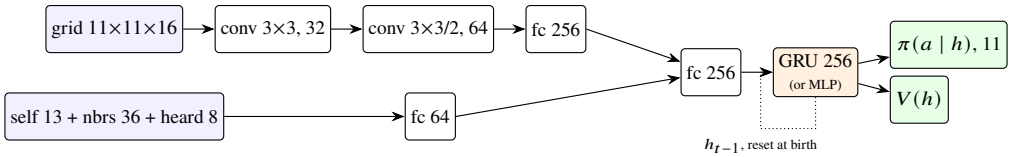


Figure 3: The shared actor-critic. The core is a GRU (memory) or an MLP (memoryless ablation).

Table 1: Held-out worlds (seeds 0–4, 2 years, 256 agents). Mean with 95% bootstrap CI over worlds; g: Hedges' g of learner minus heuristic on per-world mortality (positive = more deaths).

Policy	deaths / agent-year	KM survival at 1 yr	starvation	thirst	g vs heuristic
Random	3.83 [3.53, 4.19]	0.001	52%	34%	—
Greedy	3.31 [2.92, 3.77]	0.006	18%	61%	—
Heuristic	1.50 [1.04, 2.11]	0.214	9%	62%	—
PPO memoryless	1.50 [1.30, 1.75]	0.100	76%	10%	+0.01
PPO GRU	1.67 [1.45, 1.92]	0.055	77%	6%	+0.31
Original independent implementation (same protocol, different code)					
PPO memoryless	1.58 [1.35, 1.83]	0.101	71%	15%	+0.15
PPO GRU	1.80 [1.52, 2.10]	0.062	79%	5%	+0.50

Learner. One actor-critic network (Fig. 3; about 1.1M parameters) is shared by all 256 agents; each agent keeps its own recurrent state, reset when a newborn takes over its slot. The core is a GRU (Cho et al., 2014) (*memory*) or a feed-forward layer (*memoryless* ablation). We train with PPO (Schulman et al., 2017) and GAE (Schulman et al., 2016) on 128-tick rollouts of 256 agents, back-propagating through whole rollouts, for 600 iterations (19.7M agent-steps; Appendix Table 2). A death cuts the bootstrap. Action sampling is seeded on the CPU, so training is deterministic on CPU; the GPU (Apple MPS) path is not bit-exact.

Protocol. Training worlds use seeds $1000 + 100k + w$ (a new world every 40 iterations), validation worlds seeds 500–502 (probes and curves), and held-out test worlds seeds 0–4, never seen in training (Fig. 4). Every evaluation uses 256 agents for two years on each of the five test worlds with a fixed population (dead agents are replaced by newcomers with random genes). We report deaths per agent-year and Kaplan–Meier survival (Kaplan and Meier, 1958) with 95% bootstrap confidence intervals over worlds (Efron and Tibshirani, 1993) and Hedges' g (Hedges, 1981). Four hypotheses were registered before the first evaluation: **H1** learners beat random and greedy; **H2** learners beat the heuristic; **H3** memory reduces thirst deaths without increasing overall mortality; **H4** no social behaviour beyond what shared resources explain. "Beats" requires non-overlapping CIs.

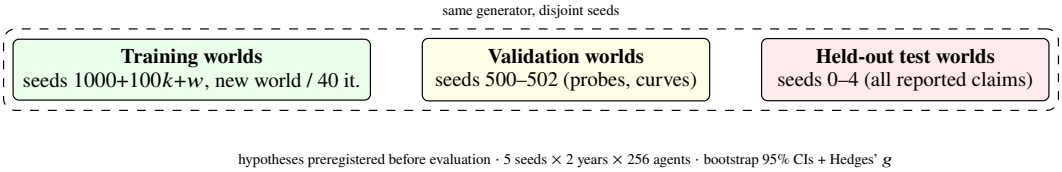
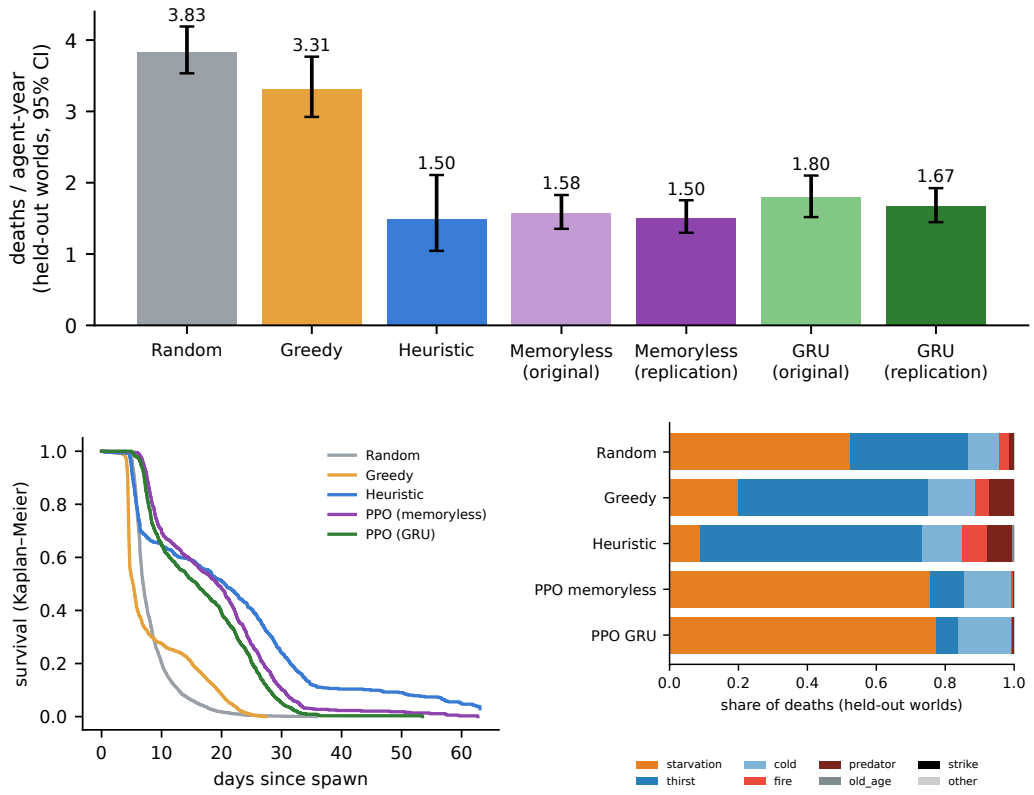


Figure 4: World splits. All reported claims use the held-out test worlds.

5 Results: survival

H1 supported, H2 not supported. Both learners more than halve mortality relative to random and greedy agents (Table 1, Fig. 5; g from −3.4 to −5.7) with non-overlapping CIs. Neither is



(a) Survival (test world 0)

(b) Causes of death (all test worlds)

Figure 5: Top: held-out mortality, including the original implementation and our replication. **Bottom:** learners trade thirst and predator deaths for starvation.

distinguishable from the heuristic at five worlds. Kaplan–Meier survival at one year is lower for learners than for the heuristic despite similar death rates, because learners' deaths are concentrated in autumn and winter (§6).

The causes of death change completely. Learners almost eliminate thirst (6–10% of deaths vs. 62% for the heuristic), fire and predator deaths, and instead die of starvation (76–77%). They have solved the problems that kill the hand-written agent and are now limited by energy.

Replication across implementations. The results were first obtained with an independent implementation of the same learner and protocol; after merging the two code bases into one trainer, we retrained both arms from scratch. All four estimates overlap (Table 1), the ordering (memoryless ≤ GRU) and the cause-of-death profile repeat, and the replication's point estimates are slightly better.

H3: memory does not help yet. With memory, thirst deaths fall (6% vs. 10%; in the original run 5% vs. 15%), but overall mortality is not lower; the memoryless agent is better on point estimates in both implementations. At 19.7M agent-steps the GRU's advantage for remembering out-of-view water is outweighed by its higher cost of learning; H3 is *not supported* at this budget.

Training dynamics. Policy entropy falls steadily from 2.40 (uniform) to 1.33–1.39 nats (Fig. 6). Training-world mortality is dominated by the world schedule and seasons and is not a usable learning curve; we have since added periodic evaluation on fixed validation worlds.

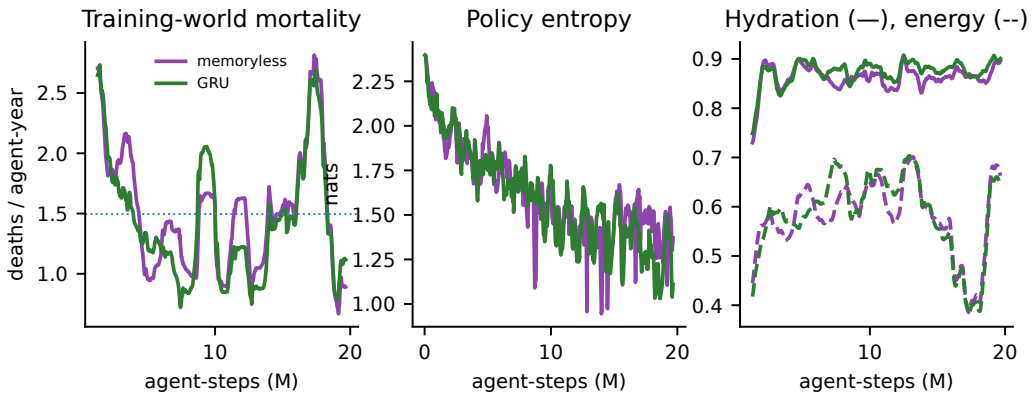
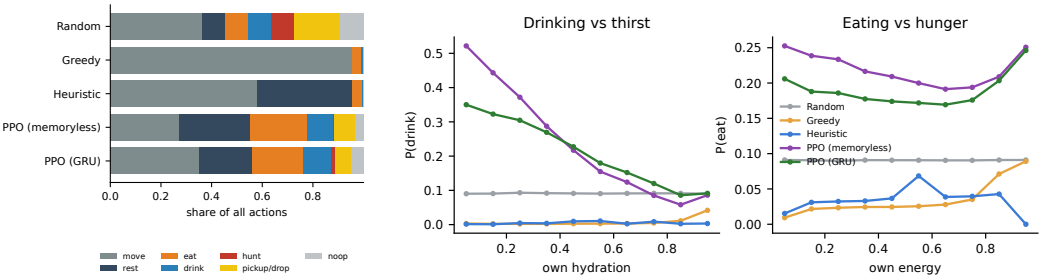


Figure 6: Training. Mortality on training worlds (40-iteration mean; dotted: heuristic's held-out rate) oscillates with world switches and seasons; entropy falls steadily; hydration rises early, energy stays seasonal.



(a) Action budget

(b) Need-conditioned choice

Figure 7: **(a)** Learners forage in many small bites. **(b)** Drinking rises smoothly with thirst; eating is U-shaped in energy. Baselines are flat (random) or near zero per tick (threshold-driven heuristic).

6 A behavioural anatomy of the learners

All analyses in this section use the complete per-tick recordings of test world 0 (two years, 256 agents continuously, ≈2M agent-ticks per policy) and compare all five policies on the same world.

Time budget: forage little and often. The heuristic moves 58% of the time and rests 38%, eating in 4% and drinking in 0.5% of ticks (large meals triggered by thresholds). Learners move *less* (27–35%), rest 21–28%, and spend 20–23% of ticks eating and 10–11% drinking (Fig. 7a): many small meals and sips.

Graded homeostatic control. The probability of drinking is a smooth, monotone function of the agent's own hydration: from 0.52 when parched to 0.09 when sated for the memoryless learner (0.35 to 0.09 with memory; Fig. 7b). Eating is U-shaped: learners eat most when hungry but also when almost full (0.25 at the top energy decile), where part of the intake is wasted against the energy cap. This explains a paradox in §7: learners *eat more* than any hand-written forager yet starve.

Riverside sedentism. Learners stay close to fresh water: median 2.0–2.2 cells, vs. 3.6 for the heuristic and 4.2 for random and greedy agents. They keep small home ranges (median radius of gyration 14–18 cells vs. 35–38 for heuristic and greedy agents; Fig. 8). Occupancy maps show them threading along the river network: a *central-place* strategy that guarantees water and lets plants near rivers regrow between visits.

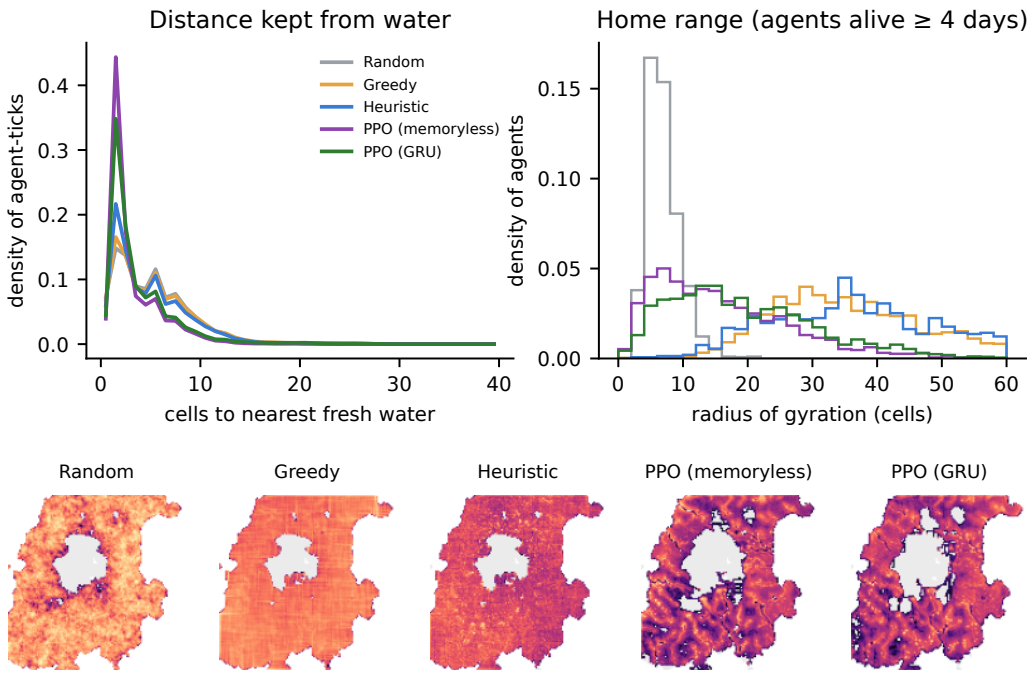


Figure 8: Space use on test world 0. **Top:** distance kept from fresh water and home-range size. **Bottom:** where each policy spent its time (log visits). Learners concentrate along rivers.

Daily and seasonal rhythms. The heuristic rests at night by construction (69% at night vs. 8% at midday). Learners show *no* night-time resting (memoryless 25% at night vs. 31% at midday; Fig. 9a), so the world's darkness (shorter sight, colder air) has not been exploited. Seasonally, learners' mean energy peaks in summer (0.75–0.79) and collapses through autumn (0.42–0.46) into winter (0.30–0.33), when plant growth stops; the heuristic, which hunts, keeps 0.69 in autumn and 0.50 in winter. Learner deaths are concentrated in autumn and winter (Fig. 9b).

What the critic tracks. The learned value estimate falls monotonically with the body's distance from its set-point ($r = -0.65$ memoryless, -0.75 GRU; Fig. 10) and is lower in autumn and winter: the critic has learned an interoceptive "well-being" signal from the reward alone. The GRU's policy becomes markedly more decisive at dawn (entropy 1.43 at night → 1.23 at 06:00), a temporal structure absent from the memoryless policy (1.44–1.50 throughout).

Individual differences. Although every agent runs the same network, individuals differ in state and history. We embed each agent that lived at least four days by its action frequencies (Fig. 11a). Policies separate cleanly; *within* a policy, k-means finds only weakly separated groups (best silhouette 0.34 memoryless, 0.41 GRU, 0.43 heuristic, against 0.11 for random agents). There is a continuum of movers and grazers, but no evidence of distinct behavioural roles.

Social structure: H4 holds. Learners are closer to each other than to random points drawn from the cells they use (nearest-neighbour ratio 0.84 vs. 0.95–0.98 for baselines; Fig. 11b), which could be read as aggregation. A stronger null that preserves each policy's own place preferences (points drawn in proportion to visits) removes the effect entirely (0.98–1.01). Learners cluster because they favour the same riverbanks, not because they are attracted to one another. Across all five test worlds (≈2,560 agent-years per policy) learners made *no* successful group kill of large prey (5 and 12 attack attempts), and food left by another agent supplied 11.4 (memoryless) and 24.5 (GRU) energy units in total, under 0.01 per agent-year against an intake of ≈6.5. That is less than the incidental scavenging

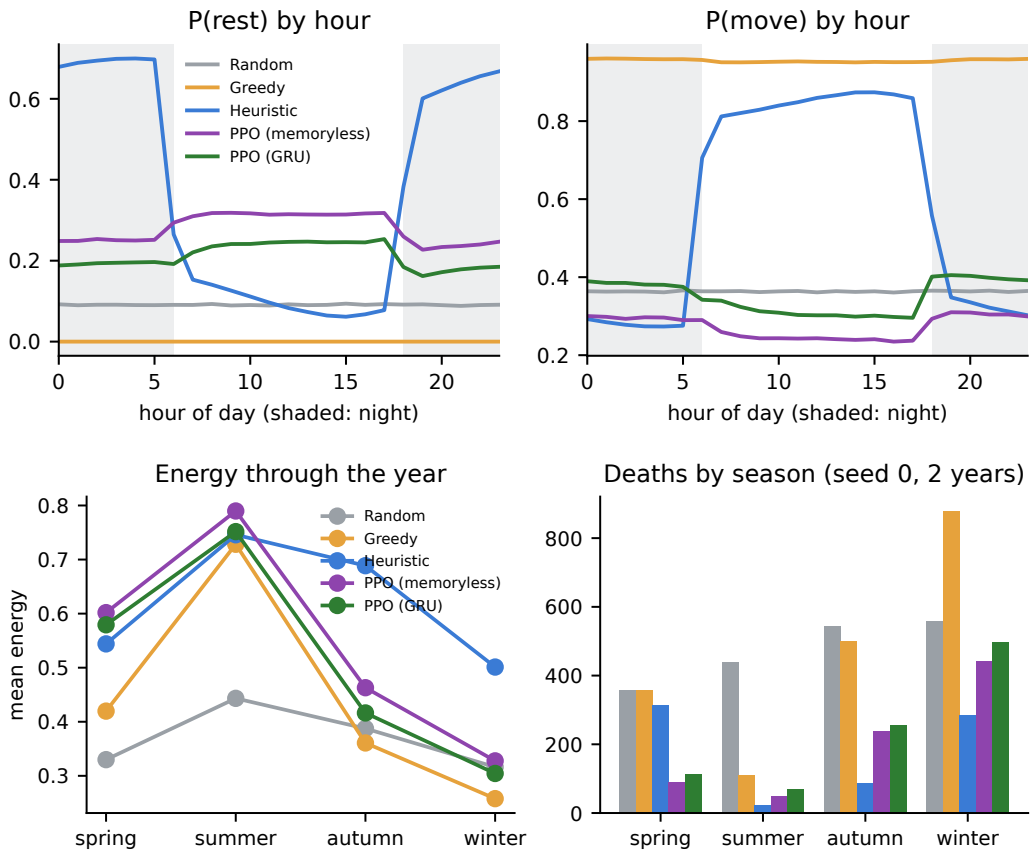


Figure 9: Top: resting and moving by hour (shaded: night). Only the hand-written heuristic rests at night. **Bottom:** energy through the year and deaths by season. Learners starve when plant growth stops.

of the rule-based heuristic, which has no social rule (65.6). We find no social behaviour at this stage, as preregistered.

7 Is starvation forced by the world? An energy-economy probe

Learners eat plants and fruit and almost never hunt. To ask whether that is a limitation of the world or of learning, we ran hand-written foragers that differ *only* in what a hungry agent seeks (identical water, shelter and danger rules) on the validation worlds (three seeds, two years), together with the learners (Fig. 12).

A pure grazer eats 3.9 energy units per agent-year, *below* basal metabolism (5.0), and dies at 3.12 per agent-year. A fruit-first forager does little better (4.3; 2.58). Adding hunting closes the gap: the hunter eats 5.2 (1.3 of it meat) and dies at 1.33, and the mixed heuristic eats 5.3 and dies at 1.24. Meat matters most in winter, when plant intake collapses (Fig. 12, right). Over the year the learners eat more than any hand-written forager (6.4–6.7, mostly plants and fruit), but the timing is wrong. They eat twice the heuristic's intake in spring and summer (2.1 and 3.1–3.2 units per agent per season, vs. 1.35 and 1.6), much of it near the energy cap where it is wasted. In winter they eat 0.12–0.15, less than a fifth of the heuristic's 0.71, because meat is the only food that does not stop growing in winter and they take almost none (0.24–0.30 per agent-year). Their deaths are 75–77% starvation, concentrated in autumn and winter. **The world rewards hunting; the learners have not found it.** Their diet is a local optimum, consistent with the credit-assignment difficulty of hunting (approach, repeated attacks, eating the carcass, with no reward until the energy arrives).

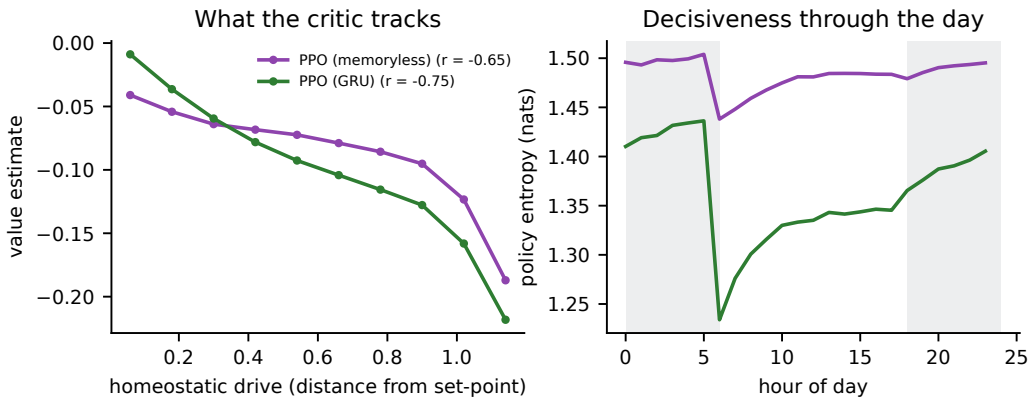
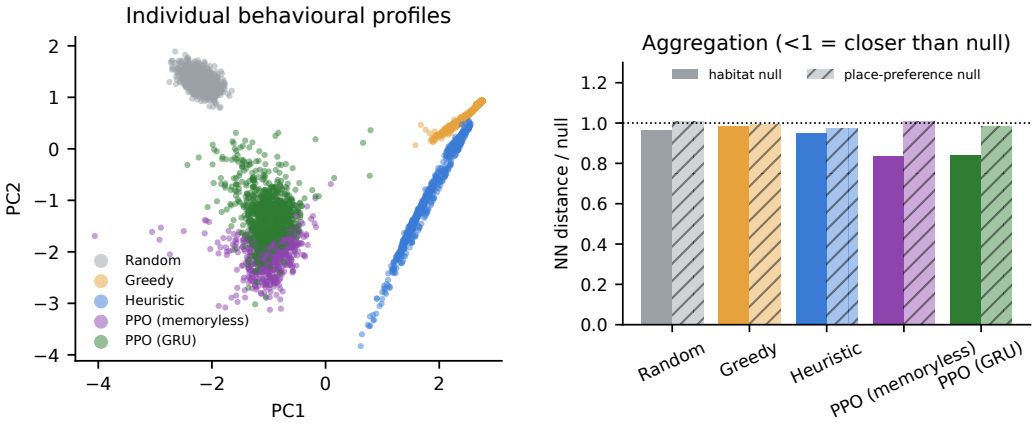


Figure 10: Learner internals. **Left:** value estimate against homeostatic drive. **Right:** policy entropy by hour; the GRU policy sharpens at dawn.



(a) Behavioural profiles (PCA)

(b) Aggregation vs. two nulls

Figure 11: **(a)** Each dot is one agent's lifetime action profile; policies separate, individuals within a learner policy form a continuum. **(b)** Apparent learner aggregation against a habitat null disappears under a place-preference null.

8 Discussion

What emerged. From a purely homeostatic reward, a single shared network learned interoceptive control (graded drinking and eating), a central-place foraging strategy anchored on rivers, and a critic that represents bodily well-being and season. Structurally, this resembles the water-tethered foraging of many mobile animals and of human foragers in arid environments. We offer this only as an analogy: it is the expected solution to a water-limited landscape, not evidence of human-like cognition.

What did not emerge. Hunting, night-time rest, food sharing, cooperative hunting and aggregation beyond shared place preference are all absent. The world makes each of them possible: group kills of large prey require two hunters, and dropped food can be taken. Their absence at 20M agent-steps sets a clear bar for what later capabilities (longer training, memory, communication, reproduction) must change. The energy probe shows the next bottleneck is ecological competence (hunting), which plausibly precedes any social use of meat. This ordering, *individual hunting before sharing*, is itself a testable prediction.

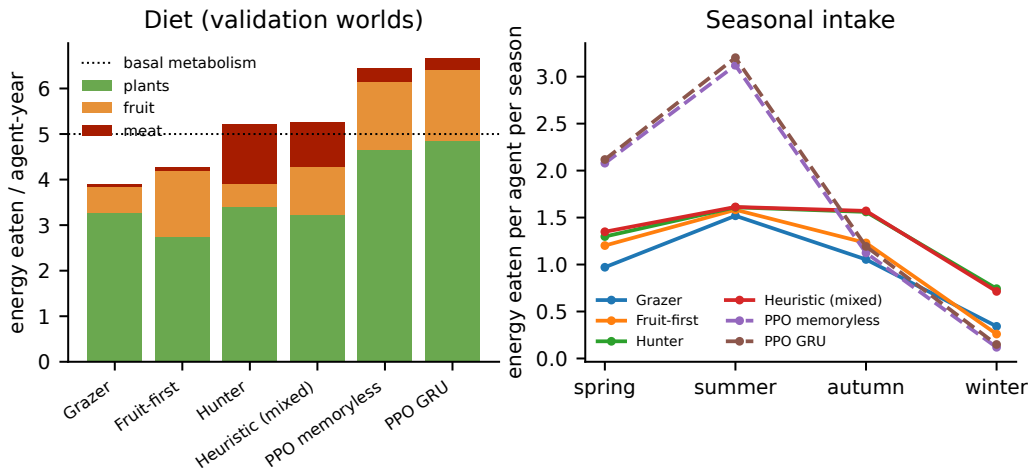


Figure 12: Energy economy on validation worlds. **Left:** energy eaten per agent-year by source; dotted line: basal metabolism alone. **Right:** seasonal intake. Plant-only grazing cannot sustain a body; adding meat closes the winter gap.

Limitations.

- **Seeds and budget:** one training seed per arm and 19.7M agent-steps, small for a 1.1M-parameter recurrent policy. H2 and H3 may change with more training.
- **Scope of the behavioural analysis:** it uses one test world. The aggregate statistics use all five.
- **Reproducibility:** MPS (GPU) arithmetic is not bit-exact, so the learner replication is statistical. The simulator itself is bit-exact.
- **Design choices:** the world's parameters (metabolism, prey energy, fauna dynamics) shape every result. We report the energy probe rather than retune the world for the learners.
- **Population model:** the fixed-population, respawn protocol has no reproduction or inheritance, so nothing here speaks to cultural or genetic accumulation.

Next steps. Longer runs (≥100M agent-steps, three seeds per arm) with periodic validation; curricula or intrinsic-motivation ablations to test whether hunting is an exploration problem; then the Stage 3 social metrics (co-feeding, food transfer, signal–context information) under the reproduction and communication switches.

Reproducibility and broader impact. Code, configurations, preregistration, evaluation files and analysis scripts are public. Every number in this paper is produced by scripts in the repository (`eval_policy.py`, `energy_economy.py`, `behaviour_analysis.py`, `paper_figures.py`). The work studies artificial agents in a simulated ecology and carries no direct societal risk. We caution against reading structural analogies with human prehistory as claims about humans.

References

B. Baker, I. Kanitscheider, T. Markov, Y. Wu, G. Powell, B. McGrew, and I. Mordatch. Emergent tool use from multi-agent autocurricula. In *ICLR*, 2020.

R. Barnes, C. Lehman, and D. Mulla. Priority-flood: an optimal depression-filling and watershed-labeling algorithm for digital elevation models. *Computers & Geosciences*, 62:117–127, 2014.

A. Bhoopchand et al. Learning few-shot imitation as cultural transmission. *Nature Communications*, 14, 2023.

K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio. Learning phrase representations using RNN encoder-decoder for statistical machine translation. In *EMNLP*, 2014.

E. A. Duéñez-Guzmán, S. Sadedin, J. X. Wang, K. R. McKee, and J. Z. Leibo. A social path to human-like artificial intelligence. *Nature Machine Intelligence*, 5:1181–1188, 2023.

B. Efron and R. J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, 1993.

J. M. Epstein and R. Axtell. *Growing Artificial Societies: Social Science from the Bottom Up*. MIT Press, 1996.

J. N. Foerster, Y. M. Assael, N. de Freitas, and S. Whiteson. Learning to communicate with deep multi-agent reinforcement learning. In *NeurIPS*, 2016.

D. Hafner. Benchmarking the spectrum of agent capabilities. In *ICLR*, 2022.

L. V. Hedges. Distribution theory for Glass's estimator of effect size and related estimators. *Journal of Educational Statistics*, 6(2):107–128, 1981.

M. B. Johanson, E. Hughes, F. Timbers, and J. Z. Leibo. Emergent bartering behaviour in multi-agent reinforcement learning. *arXiv:2205.06760*, 2022.

E. L. Kaplan and P. Meier. Nonparametric estimation from incomplete observations. *Journal of the American Statistical Association*, 53(282):457–481, 1958.

M. Keramati and B. Gutkin. Homeostatic reinforcement learning for integrating reward collection and physiological stability. *eLife*, 3:e04811, 2014.

S. K. Lam, A. Pitrou, and S. Seibert. Numba: a LLVM-based Python JIT compiler. In *LLVM-HPC Workshop*, 2015.

A. Lazaridou, A. Peysakhovich, and M. Baroni. Multi-agent cooperation and the emergence of (natural) language. In *ICLR*, 2017.

J. Z. Leibo, V. Zambaldi, M. Lanctot, J. Marecki, and T. Graepel. Multi-agent reinforcement learning in sequential social dilemmas. In *AAMAS*, 2017.

J. Z. Leibo et al. Scalable evaluation of multi-agent reinforcement learning with Melting Pot. In *ICML*, 2021.

J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein. Generative agents: interactive simulacra of human behavior. In *UIST*, 2023.

J. Perolat, J. Z. Leibo, V. Zambaldi, C. Beattie, K. Tuyls, and T. Graepel. A multi-agent reinforcement learning model of common-pool resource appropriation. In *NeurIPS*, 2017.

J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel. High-dimensional continuous control using generalized advantage estimation. In *ICLR*, 2016.

J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. *arXiv:1707.06347*, 2017.

J. Suarez, Y. Du, P. Isola, and I. Mordatch. Neural MMO: a massively multiagent game environment for training and evaluating intelligent agents. *arXiv:1903.00784*, 2019.

J. K. Terry et al. PettingZoo: Gym for multi-agent reinforcement learning. In *NeurIPS*, 2021.

L. Yaeger. Computational genetics, physiology, metabolism, neural systems, learning, vision, and behavior or PolyWorld: life in a new context. In *Artificial Life III*, 1994.

Table 2: Learner hyperparameters (identical for both arms except the core).

Agents (shared policy)	256	Rollout length	128 ticks
Iterations	600 (19.7M agent-steps)	BPTT	whole rollout
Discount γ / GAE λ	0.995 / 0.95	Epochs $\times$ minibatches	4×4 (split by agent)
Clip / entropy / value coef.	0.2 / 0.01 / 0.5	Optimiser	Adam, lr $3 \cdot 10^{-4}$, grad. clip 0.5
Hidden size	256 (GRU or MLP core)	Reward	raw homeostatic (no normalisation)
Training worlds	1000+w, new world / 40 it.	Hardware	Apple M-series laptop, MPS
Wall time	1.0 h (memoryless), 1.7 h (GRU)	Parameters	$\approx$1.1M

A Hyperparameters and environment settings

Environment. Basal metabolism 0.0013 energy per tick; plant bite 0.25 biomass $\rightarrow$ 0.03 energy; fruit 0.15; small-prey carcass 0.30 (60% kill chance per hunt action); large-prey carcass 1.8 (two attackers within six ticks); predators attack adjacent agents with probability 0.4 for 0.08 health. Fauna: energy stores, metabolism $\times$0.2 below 5 °C, births in spring to autumn only; caps and immigration floors never bind in diagnostics (5 seeds $\times$ 3 years). Full specification: `docs/WORLD.md`; all values: `configs/default.yaml`.

Diet probes. The four hand-written foragers share every rule of the heuristic (flee predators and fire, drink with hysteresis, seek shelter when cold, rest at night) and differ only in what a hungry agent seeks: plants only (grazer), fruit trees first (fruit-first), small prey first (hunter), or the heuristic's default order (mixed). The heuristic's standard path reproduces its committed baseline bit-for-bit.